

Gene Expression Profiling Using Python

PROJECT REPORT

1. Introduction

Gene expression profiling is an important technique in biotechnology and bioinformatics used to study how genes behave under different biological conditions.

In this project, gene expression data from control and treated samples were analyzed using Python. The project focused on identifying genes that were upregulated or downregulated after treatment and visualizing the expression differences using graphical methods.

It helps to understand how treatments or diseases affect cellular activity.

2. Objective

The objective of this project was to analyze gene expression differences between control and treated samples using Python. The project aimed to identify genes with increased or decreased expression and visualize the results using simple biological plots.

3. Tools and Libraries Used

- Python
 - Pandas
 - Matplotlib
 - Google Colab
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4. Dataset Information

The dataset contained gene expression values for:

- Control samples

- Treated samples

Genes included in the dataset:

- TP53
- BRCA1
- MYC
- EGFR
- GAPDH
- ACTB
- CDK1
- BCL2
- VEGFA
- TNF

The dataset was provided in CSV format.

5. Methodology

5.1. Loading the Dataset

The CSV dataset was loaded into Python using the pandas library.

5.2. Dataset Inspection

The dataset structure, rows, columns, and preview of the data were checked using pandas functions.

5.3. Calculating Average Expression

Average expression values for control and treated samples were calculated.

5.4. Comparing Gene Expression

The expression difference between treated and control samples was calculated.

5.5. Identifying Upregulated and Downregulated Genes

Genes with positive expression difference were classified as upregulated, while genes with negative expression difference were classified as downregulated.

5.6. Visualization

Bar plots were created to visualize gene expression differences and compare control and treated samples.

6. Results and Interpretation

The analysis showed that several genes had altered expression after treatment.

Upregulated Genes

The following genes showed increased expression:

- TP53
- MYC
- EGFR
- CDK1
- VEGFA
- TNF

Downregulated Genes

The following genes showed decreased expression:

- BRCA1
- BCL2

Positive expression differences indicated upregulated genes, while negative differences indicated downregulated genes. The bar plots clearly demonstrated the expression changes between control and treated conditions.

7. Conclusion

This project helped me understand how Python can be used in biological data analysis. Through this project, I understood how to load datasets, calculate gene expression differences, classify genes based on regulation, and visualize biological data using graphs.

Output Files

The following output files were generated during the project:

- Analyzed gene expression (.csv)
- Python code file (.py)
- Gene expression visualization plots (.png)
- Google Colab notebook (.ipynb)